UNIT - I

ICP [redacted]

Computer

A computer is an electronic machine that accepts information, stores it until the information is needed, processes the information according to the instructions provided by the user, and finally returns the results to the user. The computer can store and manipulate large quantities of data at very high speed, but a computer cannot think. A computer makes decisions based on simple comparisons such as one number being larger than another. Although the computer can help solve a tremendous variety of problems, it is simply a machine. It cannot solve problems on its own.

COMPUTER stands for:

- C - Common
- O - Oriented
- M - Machine
- P - Particularly
- U - Used For
- T - Trade
- E - Education And
- R – Research

Components of a Computer

Input Unit: This unit is used for entering data and programs into the computer system by the user for processing.

Storage Unit: The storage unit is used for storing data and instructions before and after processing.

Output Unit: The output unit is used for storing the result as output produced by the computer after processing.

Processing: The task of performing operations like arithmetic and logical operations is called processing. The Central Processing Unit (CPU) takes data and instructions from the storage unit and makes all sorts of calculations based on the instructions given and the type of data provided. It is then sent back to

the storage unit. CPU includes Arithmetic logic unit (ALU) and control unit (CU)

- **Arithmetic Logic Unit:** All calculations and comparisons, based on the instructions provided, are carried out within the ALU. It performs arithmetic functions like addition, subtraction, multiplication, division and also logical operations like greater than, less than and equal to etc.

Control Unit: Controlling of all operations like input, processing and output are performed by control unit. It takes care of step by step processing of all operations inside the computer.

Hardware and Software

Hardware : Computer hardware refers to the physical parts or components of a computer such as monitor, keyboard, Computer data storage, hard drive disk, mouse, printers, CPU (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that you can actually touch.

Software :A sequence of instructions given to a computer to perform a particular task is called a program. A set of programs written for a computer is called software.
For Example :C, C++, Java, Pascal, etc.

Software is divided into two types :

1. System Software
2. Application Software

System Software :A system required to execute the user program is known as system software. System software is a set of one or more programs designed to control the operation of a computer system. There are general programs written to asset humans with the help of computer by programming task.
System software consists of programs that manage the hardware resources of a computer and perform required information processing tasks. Examples of system software are Operating System, Compiler, Editor, etc.

Application Software :Application software is a set of one or more programs designed to carry out operations for specific applications. Examples of application software are MS-Word, VLC Media Player, etc.

Characteristics of Computer

The characteristics of computers that have made them so powerful and universally useful are speed, accuracy, diligence, versatility and storage capacity. Let us discuss them briefly.

Speed:Computers work at an incredible speed. A powerful computer is capable of performing about 3-4 million simple instructions per second.

Accuracy :In addition to being fast, computers are also accurate. Errors that may occur can almost always be attributed to human error (inaccurate data, poorly designed system or faulty instructions/programs written by the programmer)

Diligence :Unlike human beings, computers are highly consistent. They do not suffer from human traits of boredom and tiredness resulting in lack of concentration. Computers, therefore, are better than human beings in performing voluminous and repetitive jobs.

Versatility :Computers are versatile machines and are capable of performing any task as long as it can be broken down into a series of logical steps. The presence of computers can be seen in almost every sphere – Railway/Air reservation, Banks, Hotels, Weather forecasting and many more.

Storage Capacity:Today's computers can store large volumes of data. A piece of information once recorded (or stored) in the computer, can never be forgotten and can be retrieved almost instantaneously.

Classification of Computers

1. Mainframe Computers
2. Minicomputers
3. Microcomputers
4. Supercomputers

Mainframe computers are very large, often filling an entire room. They can store enormous of information, can perform many tasks at the same time, can communicate with many users at the same time, and are very expensive. . The price of a mainframe computer frequently runs into the millions of dollars. Mainframe computers usually have many terminals connected to them. These terminals look like small computers but they are only devices used to send and receive information from the actual computer using wires. Terminals can be

located in the same room with the mainframe computer, but they can also be in different rooms, buildings, or cities. Large businesses, government agencies, and universities usually use this type of computer.

Minicomputers are much smaller than mainframe computers and they are also much less expensive. The cost of these computers can vary from a few thousand dollars to several hundred thousand dollars. They possess most of the features found on mainframe computers, but on a more limited scale. They can still have many terminals, but not as many as the mainframes. They can store a tremendous amount of information, but again usually not as much as the mainframe. Medium and small businesses typically use these computers.

Microcomputers are the types of computers we are using in your classes at Floyd College. These computers are usually divided into desktop models and laptop models. They are terribly limited in what they can do when compared to the larger models discussed above because they can only be used by one person at a time, they are much slower than the larger computers, and they can not store nearly as much information, but they are excellent when used in small businesses, homes, and school classrooms. These computers are inexpensive and easy to use. They have become an indispensable part of modern life.

Operating System

The operating system is the most important program that runs on a computer.An **operating system** or **OS** is a software program that enables the computerhardware to communicate and operate with the computer software. Without a computer operating system, a computer and software programs would be useless. Every general-purpose computer must have an operating system to run other programs. Operating systems perform basic tasks, such as recognizing inputfrom the keyboard, sending output to the display screen, keeping track of filesand directories on the disk, and controlling peripheral devices such as disk drives and printers.
For large systems, the operating system has even greater responsibilities and powers. It is like a traffic cop -- it makes sure that different programs andusers running at the same time do not interfere with each other. The operating system is also responsible for *security*, ensuring that unauthorized users do not access the system. Examples of Operating System are : Windows 98/2000/2003/2005/2008, Linux, Unix, DOS, etc.

Compiler

A **compiler** is a computer program (or set of programs) that transforms source code written in a programming language(the *source language*) into another

computer language (the *target language*, often having a binary form known as *object code*). The most common reason for wanting to transform source code is to create an executable program.

A compiler is a piece of code that translates the high level language into machine language. When a user writes a code in a high level language such as C and wants it to execute, a specific compiler which is designed for C is used before it will be executed. The compiler scans the entire program first and then translates it into machine code which will be executed by the computer processor and the corresponding tasks will be performed.

Interpreter

Interpreters are not much different than compilers. They also convert the high level language into machine readable binary equivalents. Each time when an interpreter gets a high level language code to be executed, it converts the code into an intermediate code before converting it into the machine code. Each part of the code is interpreted and then execute separately in a sequence and an error is found in a part of the code it will stop the interpretation of the code without translating the next set of the codes.

Difference between Compiler and Interpreter

1. The interpreter takes one statement then translates it and executes it and then takes another statement. While the compiler translates the entire program in one go and then executes it.
2. Compiler generates the error report after the translation of the entire page while an interpreter will stop the translation after it gets the first error.
3. Compiler takes a larger amount of time in analyzing and processing the high level language code comparatively interpreter takes lesser time in the same process.
4. Besides the processing and analyzing time the overall execution time of a code is faster for compiler relative to the interpreter.

Assembler

An assembler is a computer program which translates from assembly language to an object file or machine language format.

Linker

In computer science, a **linker** or **link editor** is a computer program that takes one or more object files generated by a compiler and combines them into a single executable program.

Loader

In computing, a **loader** is the part of an operating system that is responsible for loading programs. It is one of the essential stages in the process of starting a program, as it places programs into memory and prepares them for execution. Loading a program involves reading the contents of executable file, the file containing the program text, into memory, and then carrying out other required preparatory tasks to prepare the executable for running. Once loading is complete, the operating system starts the program by passing control to the loaded program code.

Bit

A **bit** is the basic unit of information in computing and telecommunications. A bit can have only two values: either 1 or 0. A bit can be implemented in hardware by means of a two state device. Eight bits form a byte.

In computing, a bit can be defined as a variable or computed quantity that can have only two possible values. These two values are often interpreted asbinary digits and are usually denoted by the numerical digits 0 and 1. The two values can also be interpreted as logical values (*true/false*, *yes/no*), algebraic signs (+/−), activation states (*on/off*), or any other two-valued attribute. The correspondence between these values and the physical states of the underlying storage or device is a matter of convention, and different assignments may be used even within the same device or program. The length of a binary number may be referred to as its "bit-length".

Byte

The combination of eight bits is known as byte or in other words, we can say that eight bits form a byte.

Booting Process

In short booting refers to the process of loading the operating system (such as Windows) to the RAM (working memory) of the computer, which otherwise is stored in the hard disk (secondary memory). Computer cannot access anything which is stored in the secondary memory unless it is loaded to the RAM.

In computing, **booting** (also known as **booting up**) is the initial set of operations that a computer system performs when electrical power to the CPU is switched on. The process begins when a computer is turned on for the first time or is re-energized after being turned off, and ends when the computer is ready to perform its normal operations. On modern general purpose computers, this can take tens of seconds and typically involves performing a power-on self-test, locating and initializing peripheral devices, and then finding, loading and starting an operating system. Many computer systems also allow these operations to be initiated by a software command without cycling power, in what is known as a soft reboot, though some of the initial operations might be skipped on a soft reboot. A **boot loader** is a computer program that loads the main operating system or runtime environmentfor the computer after completion of the self-tests.

The computer term *boot* is short for *bootstrap* or *bootstrap load* and derives from the phrase *to pull oneself up by one's bootstraps*. The usage calls attention to the paradox that a computer cannot run without first loading software but some software must run before any software can be loaded.[4] Early computers used a variety of ad-hoc methods to get a small program into memory to solve this problem. The invention of integrated circuit read-only memory (ROM) of various types solved this paradox by allowing computers to be shipped with a start up program that could not be erased. Growth in the capacity of ROM has allowed ever more elaborate start up procedures to be implemented.

On general purpose computers, the boot process begins with the execution of an initial program stored in boot ROMs or read in another fashion. In some older computers, the initial program might have been the application to run, if no operating system was used, or the operating system. In other computers, the initial program is a boot loader that may then load into random-access memory (RAM), from nonvolatile secondary storage (such as a hard disk drive) or, in some older computers, from a medium such as punched cards,punched tape, or magnetic tape in older computers, the binary code of an operating system or runtime environment and then execute it. If the boot loader is limited in its size and capabilities, it may, instead, load a larger and more capable secondary boot loader, which would then load the operating system or runtime environment. Some embedded systemsdo not require a noticeable boot sequence to begin functioning and when turned on may simply run operational programs that are stored in ROM.